

D11. Executive Summary

Overview Summary

FlowCore is a compact inbound warehouse commissioning simulation. It demonstrates how a small warehouse process can be configured, executed, stored, validated, and explained through multiple technical layers.

The selected scenario processes one inbound receipt at DOCK-01 and stores three pallets in controlled warehouse locations. Each pallet follows the lifecycle RECEIVED -> ASSIGNED -> STORED.

Technical Outcome

The project uses a layered structure. Bash runs the local workflow and generates CSV evidence. Oracle SQL stores and queries the warehouse data. PL/SQL provides one simple function for available location capacity. C++ applies a priority placement decision rule using the generated runtime evidence.

Implementation Scope

The implementation is intentionally limited to a clear commissioning scenario rather than a large warehouse management system. The scope includes receipt creation, location setup, pallet storage, status history, validation queries, a PL/SQL capacity function, and a C++ priority placement rule. This keeps the project realistic, testable, and easy to explain within a short technical presentation.

Validated Result

- **Stored pallets:** 3 final pallet records in STORED status.
- **Status events:** 9 lifecycle events, covering RECEIVED, ASSIGNED, and STORED for each pallet.
- **Capacity violations:** 0, meaning no location exceeds its configured capacity.
- **Priority pallets checked by C++:** 2 high-quantity pallets.
- **C++ priority decision:** passed, meaning both high-quantity pallets were assigned to active priority locations.

Why The Project Is Valuable

- It shows an end-to-end warehouse process from input data to final proof.
- It produces auditable evidence files instead of relying only on screenshots.
- It separates responsibilities across Bash, SQL, PL/SQL, and C++.
- It stays small enough to explain clearly during a short technical review.
- It demonstrates that the team can interpret a case study, implement it, validate it, and explain the result.